
PropCRM

Lead Management System for Real Estate Agents

PRODUCT REQUIREMENTS DOCUMENT

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VERSION	DATE	AUTHOR	DESCRIPTION
0.1	Jul 5, 2026	Engineering	Initial draft — requirements gathering
0.9	Jul 6, 2026	Engineering	Finalized functional requirements, added NFRs
1.0	Jul 7, 2026	Engineering	Complete PRD — submitted for approval

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1. Executive Summary

PropCRM is a lightweight, web-based Customer Relationship Management system designed specifically for independent real estate agents operating in the Indian market. The system captures property inquiry leads through a public-facing form, stores them in Google Sheets, and provides a secure agent dashboard for lead management, communication, and analytics.

The product eliminates the need for expensive CRM subscriptions by leveraging free-tier services — Google Sheets for data storage, Google Apps Script for backend logic, GitHub Pages for hosting, and Brevo for transactional email delivery.

Key Value Proposition: A zero-cost, fully functional CRM that an individual real estate agent can deploy in under 30 minutes, with automated lead notifications, email follow-ups, and visual analytics — without any recurring subscription fees.

2. Problem Statement

Independent real estate agents in India face several challenges in managing their leads:

- **Manual lead tracking:** Leads arrive via phone calls, WhatsApp messages, website forms, and social media — scattered across platforms with no unified view.
- **No follow-up system:** Without automated reminders, agents frequently miss or delay following up on leads, resulting in lost conversions.
- **No data visibility:** Agents lack insight into which price brackets or localities generate the most interest, preventing them from optimizing their marketing spend and listings.
- **High CRM costs:** Enterprise CRM tools (Salesforce, HubSpot, Zoho) are priced for teams and are overkill for a single agent handling 50–500 leads per month.

3. Goals & Objectives

3.1 Primary Goals

ID	GOAL	MEASURABLE TARGET
G1	Centralize all property inquiry leads in one system	100% of web leads captured in Sheets within 2 seconds
G2	Automate lead notification to the agent	Agent receives email within 30 seconds of form submission
G3	Reduce lead response time	Reminder sent if lead is uncontacted for > 1 hour
G4	Provide actionable analytics	Real-time price bracket & locality distribution charts
G5	Zero operational cost	\$0/month recurring cost using free-tier services

3.2 Secondary Goals

- Provide a professional, branded lead capture form that builds credibility with prospects.
- Enable the agent to send professional follow-up emails directly from the dashboard.
- Create a scalable architecture that can later support multiple agents if the business grows.

4. Target Users & Personas

Persona 1: Property Seeker (Customer)

Role	End consumer looking to buy/rent property
Demographics	Age 25–55, urban Indian, smartphone-first, may use desktop
Goal	Submit a property inquiry quickly and receive a response from the agent
Pain Points	Lengthy forms, no confirmation of submission, slow agent response
Expectations	Simple form, instant confirmation, agent follow-up within hours

Persona 2: Real Estate Agent

Role	Independent real estate agent managing property sales/rentals
Demographics	Age 28–50, moderate tech literacy, manages 50–500 leads/month
Goal	Track all leads, prioritize follow-ups, understand market demand
Pain Points	Leads slip through cracks, no analytics, expensive CRM tools
Expectations	Single dashboard to view, manage, and contact all leads with visual insights

5. Scope Definition

5.1 In Scope (v1.0)

CATEGORY	FEATURE
Lead Capture	Public web form with fields: Name, Email, Property Area (locality), Property Price
Data Storage	All leads stored in Google Sheets with auto-generated ID, timestamp, and price bracket
Agent Dashboard	PIN-protected dashboard with leads table, stats cards, and analytics charts
Lead Management	Mark leads as contacted (checkbox), send follow-up email to lead
Email — Acknowledgment	Automatic email to customer upon form submission
Email — Agent Notification	Immediate email to agent when a new lead is submitted
Email — Reminder	Automated reminder email to agent for uncontacted leads (configurable hours)
Analytics	Price bracket distribution (doughnut chart), area/locality distribution (bar chart)
Configuration	Agent name, email, Brevo API key, PIN, and reminder hours stored in Google Sheets

5.2 Out of Scope (Future Versions)

FEATURE	REASON FOR EXCLUSION	TARGET VERSION
Multi-agent support	v1 is designed for a single agent	v2.0
WhatsApp integration	Requires WhatsApp Business API setup	v1.5
Lead stages/pipeline	Adds complexity; v1 focuses on contact tracking	v2.0
Lead source tracking (UTM)	Enhancement after core is validated	v1.5
User authentication (OAuth)	Overkill for single agent; PIN is sufficient	v2.0
Mobile native app	Responsive web is sufficient for v1	v3.0
CSV/Excel export	Agent can access Google Sheets directly	v1.5
Duplicate lead detection	Enhancement after core flow is stable	v1.5

6. Functional Requirements

FR-1: Lead Capture Form

REQ ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-1.1	Form shall contain fields: Full Name, Email Address, Property Area (locality), Property Price (₹)	MUST HAVE	All four fields render correctly on desktop and mobile
FR-1.2	All fields are mandatory. Form shall not submit with any empty field.	MUST HAVE	Submit button is disabled or shows error for incomplete forms
FR-1.3	Name must be at least 2 characters, alphabetic with spaces allowed	MUST HAVE	Inline error message shown for invalid name
FR-1.4	Email must be a valid email format (RFC 5322 compliant regex)	MUST HAVE	Inline error message shown for invalid email
FR-1.5	Price must be a positive number greater than zero	MUST HAVE	Non-numeric or negative values rejected with error
FR-1.6	Form shall display a loading spinner during submission	SHOULD HAVE	Submit button shows spinner, prevents double-click
FR-1.7	On successful submission, form shall display a success toast and reset	MUST HAVE	Success message visible for 5 seconds, form fields cleared
FR-1.8	On failed submission, form shall display an error toast with retry option	MUST HAVE	Error message visible, form data preserved for retry
FR-1.9	Form design shall use glassmorphism effect with cream	SHOULD HAVE	Visual review approved by stakeholder

REQ ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
	and green color palette		

FR-2: Data Storage (Google Sheets)

REQ ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-2.1	Each lead shall be stored as a row in the "Leads" sheet with columns: ID, Name, Email, Area, Price, Price Bracket, Submitted At, Contacted, Contacted At, Email Sent To Lead, Reminder Sent To Agent	MUST HAVE	New row appears in Sheets within 3 seconds of form submission
FR-2.2	Lead ID shall be a unique auto-generated identifier (UUID v4 or timestamp-based)	MUST HAVE	No duplicate IDs across all leads
FR-2.3	Price bracket shall be auto-calculated: Under 25L, 25L–50L, 50L–1Cr, 1Cr–2Cr, 2Cr–5Cr, 5Cr+	MUST HAVE	Correct bracket assigned for boundary values (e.g., ₹25,00,000 → "25L–50L")
FR-2.4	Agent configuration shall be stored in a separate "AgentConfig" sheet	MUST HAVE	Agent name, email, Brevo key, PIN, and reminder hours readable by the system
FR-2.5	Server-side validation shall reject malformed data before writing to Sheets	MUST HAVE	Invalid submissions return HTTP 400 with error description

FR-3: Agent Dashboard

REQ ID	REQUIREMENT	PRIORITY	ACCEPTANCE CRITERIA
FR-3.1	Dashboard shall be protected by a PIN/password login screen	MUST HAVE	Dashboard content hidden until correct PIN entered
FR-3.2	Dashboard shall display summary stats: Total Leads, Contacted %, Uncontacted Count, Average Property Price	MUST HAVE	Stats update on page load and on manual refresh
FR-3.3	Price distribution shall be visualized as a doughnut/bar chart with percentage labels	MUST HAVE	Chart segments match the 6 defined price brackets with green gradient coloring
FR-3.4	Area/locality distribution shall be visualized as a horizontal bar chart	MUST HAVE	Top N localities displayed with lead count
FR-3.5	Chart colors shall follow green gradient: higher concentration → darker green	SHOULD HAVE	Visual review confirms gradient is clear and readable
FR-3.6	Leads table shall display all lead data with columns: Name, Email, Area, Price, Date, Contacted, Actions	MUST HAVE	Table is scrollable, sortable by date, and matches Sheets data
FR-3.7	Each row shall have a "Contacted" checkbox that updates the lead status in real-time	MUST HAVE	Checkbox toggle updates Google Sheet within 3 seconds
FR-3.8	Each row shall have a "Send Email" button to send a follow-up email to the lead	MUST HAVE	Email sent via Brevo, confirmation shown to agent
FR-3.9	Dashboard shall have a manual "Refresh" button to re-fetch latest data	SHOULD HAVE	Data refreshes without full page reload

FR-4: Email System (via Brevo API)

REQ ID	REQUIREMENT	PRIORITY	TRIGGER	ACCEPTANCE CRITERIA
FR-4.1	Send acknowledgment email to lead upon form submission	MUST HAVE	Form submit → Apps Script	Lead receives email within 60 seconds of submission
FR-4.2	Send new lead notification email to agent upon form submission	MUST HAVE	Form submit → Apps Script	Agent receives email with lead details within 60 seconds
FR-4.3	Send reminder email to agent for leads uncontacted beyond configured hours	MUST HAVE	Scheduled trigger (every 1 hour)	Reminder sent only once per lead; "reminderSent" flag set to TRUE
FR-4.4	Agent can send a follow-up email to any lead from the dashboard	MUST HAVE	Agent clicks "Send Email" button	Fixed template email sent, button shows success state
FR-4.5	All emails shall be sent via Brevo SMTP API (server-side only)	MUST HAVE	All email actions	API key never exposed in frontend code

7. Non-Functional Requirements

7.1 Performance

REQ ID	REQUIREMENT	TARGET
NFR-1.1	Form page load time (First Contentful Paint)	< 2 seconds on 4G
NFR-1.2	Form submission response time	< 3 seconds end-to-end
NFR-1.3	Dashboard data load time	< 5 seconds for up to 1,000 leads
NFR-1.4	Chart rendering time	< 1 second after data is loaded

7.2 Security

REQ ID	REQUIREMENT	IMPLEMENTATION
NFR-2.1	Brevo API key must never be exposed in frontend code	Key stored in Google Sheets; accessed only by Apps Script server-side
NFR-2.2	Dashboard access must be gated	PIN-based authentication verified server-side
NFR-2.3	All API communication over HTTPS	GitHub Pages and Apps Script enforce HTTPS by default
NFR-2.4	Input sanitization on all user-submitted data	Server-side sanitization in Apps Script before writing to Sheets

7.3 Scalability

REQ ID	REQUIREMENT	LIMIT
NFR-3.1	System shall support up to 10,000 leads in Google Sheets	Google Sheets supports 10M cells
NFR-3.2	System shall handle up to 30 concurrent form submissions	Apps Script quota: 30 simultaneous executions

7.4 Availability & Reliability

REQ ID	REQUIREMENT	NOTE
NFR-4.1	Frontend availability: 99.9% (GitHub Pages SLA)	No self-managed infrastructure
NFR-4.2	Backend availability: Dependent on Google Apps Script uptime	Google provides 99.9% SLA for Workspace
NFR-4.3	Email delivery: Dependent on Brevo SLA	Brevo free tier: 300 emails/day

7.5 Compatibility

REQ ID	REQUIREMENT
NFR-5.1	Form page: Compatible with Chrome, Firefox, Safari, Edge (latest 2 versions)
NFR-5.2	Form page: Responsive design — mobile (360px+), tablet (768px+), desktop (1024px+)
NFR-5.3	Dashboard: Optimized for desktop and tablet (768px+)

8. Technical Architecture

8.1 High-Level Architecture

LAYER	TECHNOLOGY	HOSTING	COST
Frontend (Static)	HTML5, CSS3, Vanilla JavaScript, Chart.js	GitHub Pages	Free
Backend (API)	Google Apps Script (Web App)	Google Cloud (managed)	Free
Database	Google Sheets	Google Drive	Free
Email Service	Brevo v3 REST API	Brevo Cloud	Free (300/day)

8.2 API Contract

POST Endpoint: doPost(e)

All requests are sent as POST with a JSON body. The action field determines routing.

ACTION	PAYLOAD	RESPONSE
submitLead	{ name, email, area, price }	{ success: true, id: "..." } or { success: false, errors: [...] }
updateContacted	{ id, contacted: true/false, pin }	{ success: true }
sendEmailToLead	{ id, pin }	{ success: true }
verifyPin	{ pin }	{ success: true, agentName: "..." }

GET Endpoint: doGet(e)

PARAMETER	RESPONSE
? action=getLeads&pin=XXXX	JSON array of all leads + computed stats (totals, percentages, chart data)

9. Dependencies & Integrations

DEPENDENCY	TYPE	PURPOSE	FALLBACK
Google Sheets API	Core	Data storage and retrieval	None — core dependency
Google Apps Script	Core	Backend logic, API endpoints	None — core dependency
GitHub Pages	Core	Static frontend hosting	Netlify / Vercel (free tier)
Brevo API v3	External	Transactional email delivery	Apps Script MailApp (limited)
Chart.js v4.x	Library (CDN)	Data visualization	Manual HTML/CSS charts
Google Fonts (Inter)	Library (CDN)	Typography	System font stack

10. Success Metrics & KPIs

METRIC	TARGET	MEASUREMENT METHOD
Lead capture rate	100% of form submissions stored in Sheets	Compare form submissions vs. Sheets row count
Agent notification speed	< 60 seconds from form submit to agent email	Timestamp comparison
Lead response time	< 2 hours average (aided by reminders)	submittedAt vs. contactedAt delta
Email delivery rate	> 95% delivery (Brevo SLA)	Brevo dashboard analytics

METRIC	TARGET	MEASUREMENT METHOD
Dashboard uptime	99.9%	GitHub Pages status
Agent satisfaction	Qualitative — positive feedback	Agent interview after 2 weeks

11. Risk Assessment

RISK	PROBABILITY	IMPACT	MITIGATION
Google Apps Script rate limits exceeded	LOW	HIGH	Quota is 20,000 URL fetch calls/day — sufficient for < 500 leads/day. Monitor usage.
Brevo free tier email limit hit (300/day)	MEDIUM	MEDIUM	Each lead triggers max 3 emails (ack + notify + reminder). Supports ~100 leads/day. Upgrade Brevo plan if exceeded.
Google Sheets performance degrades at scale	LOW	MEDIUM	Archive old leads periodically. Sheets supports 10M cells. Consider migration to Firebase at 5,000+ leads.
PIN-based auth is weak security	MEDIUM	LOW	Acceptable for v1 single-agent use. Dashboard URL is not indexed by search engines. Upgrade to OAuth in v2.
CORS issues between GitHub Pages and Apps Script	LOW	HIGH	Apps Script Web Apps handle CORS natively when deployed as "Anyone can access." Tested during development.

12. Timeline & Milestones

PHASE	DURATION	DELIVERABLES	STATUS
Phase 1: Documentation	Day 1	PRD, Data Flow, User Flows	Complete
Phase 2: Google Sheets Setup	Day 1	Sheets structure, AgentConfig populated	Pending
Phase 3: Backend (Apps Script)	Day 1-2	All API endpoints, email functions, validation	Pending
Phase 4: Lead Form (Frontend)	Day 2	Form page with validation, glassmorphism design	Pending
Phase 5: Agent Dashboard (Frontend)	Day 2-3	Dashboard with table, charts, email actions	Pending
Phase 6: Integration Testing	Day 3	End-to-end testing of all flows	Pending
Phase 7: Deployment	Day 3	GitHub Pages live, Apps Script deployed, triggers set	Pending

13. Appendix

A. Price Bracket Mapping

PRICE RANGE (₹)	BACKET LABEL
0 – 24,99,999	Under 25L
25,00,000 – 49,99,999	25L – 50L
50,00,000 – 99,99,999	50L – 1Cr

PRICE RANGE (₹)	BRACKET LABEL
1,00,00,000 – 1,99,99,999	1Cr – 2Cr
2,00,00,000 – 4,99,99,999	2Cr – 5Cr
5,00,00,000+	5Cr+

B. Email Template Content

TEMPLATE	SUBJECT LINE	RECIPIENT
Lead Acknowledgment	"Thank you for your interest, {name}!"	Customer (lead)
New Lead Notification	"🏠 New Lead: {name} — ₹{price} in {area}"	Agent
Uncontacted Reminder	"🕒 Reminder: {name} hasn't been contacted yet"	Agent
Agent Follow-up	"Regarding your property inquiry — {agentName}"	Customer (lead)

C. Glossary

TERM	DEFINITION
Lead	A potential customer who has submitted the property inquiry form
Contacted	A lead that the agent has followed up with (marked via checkbox)
Price Bracket	A predefined range category for property prices in the Indian market
Brevo	Third-party transactional email service (formerly Sendinblue)
Apps Script	Google's JavaScript-based platform for extending Google Workspace
PIN	Personal Identification Number used for dashboard authentication

